JavaScript Lab — Practical 9

Course: JavaScript Lab / Web Technology
Practical No.: 09
Topic: Web Storage API (`localStorage` & `sessionStorage`) & DOM Manipulation
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📌 Aim & Objectives

1. Understand the concepts and implementation of the HTML5 **Web Storage API**.
2. Learn the differences between `localStorage` (persistent client-side storage) and `sessionStorage` (session-scoped client-side storage).
3. Implement theme preference persistence across browser reloads.
4. Implement a case study (**Seminar Schedule Planner**) that utilizes HTML table structures (`rowspan` , `colspan`), event listeners, interactive alerts, active selection states, and structured JSON data caching in Web Storage.

📂 Project Structure

```
pract 9/
├── README.md                    # Comprehensive documentation of Practical 9
├── PRACTICAL/                   # Core Practical 9 implementation
│   └── index.html               # Theme preference switcher (Light/Dark/Clear) using Web
Storage
└── CASE STUDY/                  # Extended Case Study application
    ├── index.html               # Seminar Schedule Planner UI with complex table &
storage display
    ├── style.css                # Modern UI styles, color-coded time badges, dark theme
support
    └── script.js                # Event listeners, Web Storage operations, JSON parsing,
UI renderer
```

🔬 Theoretical Background

Web Storage API Overview

HTML5 Web Storage enables web applications to store key-value data directly inside the user's browser securely and efficiently without sending data to the server on every HTTP request (unlike traditional HTTP cookies).

Feature	`localStorage`	`sessionStorage`	Cookies
Persistence	Permanent until explicitly cleared	Cleared when browser tab/window closes	Set via expiration date
Capacity	~5MB - 10MB per origin	~5MB per origin	~4KB
Scope	Shared across all tabs & windows of origin	Isolated to the specific browser tab	Sent with every HTTP request
Access	Client-side JavaScript	Client-side JavaScript	Server & Client-side

Key Storage Methods Used

- `Storage.setItem(key, value)` : Saves an item with a specified string key and value.
- `Storage.getItem(key)` : Retrieves the value associated with the specified key (returns `null` if not present).
- `Storage.removeItem(key)` : Deletes the specified key-value pair from storage.
- `JSON.stringify(obj)` & `JSON.parse(str)` : Serializes objects to strings and deserializes strings back to objects for complex data storage.

💻 Section 1: Core Practical (`PRACTICAL/index.html`)

Objective

Create a theme preference manager allowing users to select between Light and Dark themes, persist the choice using `localStorage` and `sessionStorage`, and restore the theme automatically on page reload.

Key Features

- **Light / Dark Theme Switching:** Toggles the `.dark` CSS class dynamically on the `<body>` element.
- **Persistent State:** Saves the selected theme in `localStorage.setItem('theme', theme)` so preferences persist even if the browser is closed and reopened.
- **Session Tracking:** Synchronously saves the active session preference using `sessionStorage.setItem('currentTheme', theme)` .
- **Reset Option:** `clearTheme()` deletes saved preferences via `localStorage.removeItem()` and reverts the UI to default.
- **Auto-Restore on Load:** `window.onload` checks `localStorage.getItem('theme')` and automatically applies the user's preferred visual theme.

🏢 Section 2: Case Study (`CASE STUDY/`)

Application: Seminar Schedule Planner

An interactive schedule dashboard for technical seminars featuring XML, XPath, and XSL technologies.

Key Components & Implementations

1. Advanced HTML Table Design (`index.html`)

- Utilizes multi-level nested headers (`th` with `rowspan="3"` and `colspan="3"`).
- Uses `rowspan` for merged multi-hour slots (e.g., Monday 8:00 a.m. to 5:00 p.m., Tuesday XPath session).
- Employs semantic `data-*` attributes (`data-day` , `data-begin` , `data-end` , `data-topic`) to store seminar metadata cleanly inside DOM nodes.

2. Interactive Event Handling (`script.js`)

- **Click Listener on Topic Cells:** Clicking any topic cell retrieves schedule data directly from `cell.getAttribute()` .
- **Alert Dialog:** Displays formatted seminar information in an alert prompt: ```text Seminar Schedule Details:

Day : Monday Time : 8:00 a.m. to 5:00 p.m. Topic : Introduction to XML

Saved to Local Storage! ```

3. Dual Storage Architecture

- **`localStorage` (`selectedSeminar`):**
- Serializes full seminar objects into JSON strings: `javascript const scheduleData = { day: day, begin: begin, end: end, topic: topic, savedAt: new Date().toLocaleString() }; localStorage.setItem('selectedSeminar', JSON.stringify(scheduleData));`
- Displays selected seminar details in the "Storage Info" dashboard card.
- Highlights the selected seminar cell on the table with an active outline (`.active-selected`).
- **`sessionStorage` (`viewedTopics`):**
- Keeps an array of unique seminar topics clicked during the active browsing session.
- Automatically resets when the browser tab is closed.
- **Theme Persistence (`themePreference` & `sessionTheme`):**
- Dedicated Light / Dark theme controls with independent persistence.

4. User Controls & Actions

- **Clear Saved Schedule:** Clears `selectedSeminar` from `localStorage` and resets the dashboard to its empty state.
- **Reload from Storage:** Reads and re-renders the latest state directly from storage.
- **Reset Theme:** Clears saved theme preferences and restores default styling.

📊 Storage Keys Reference

Storage Engine	Key Name	Data Type	Description
`localStorage`	`selectedSeminar`	JSON Object (stringified)	Currently selected seminar details and timestamp
`localStorage`	`themePreference`	String (`"light"` | `"dark"`)	Global persisted UI theme preference
`sessionStorage`	`viewedTopics`	JSON Array (stringified)	List of seminar topics accessed in current session
`sessionStorage`	`sessionTheme`	String (`"light"` | `"dark"`)	Active session theme tracking
`localStorage` (Practical)	`theme`	String (`"light"` | `"dark"`)	Saved theme for core practical
`sessionStorage` (Practical)	`currentTheme`	String (`"light"` | `"dark"`)	Session theme for core practical

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🚀 How to Run

1. **Clone or Open Folder:** Open the folder `pract 9` in Visual Studio Code or any modern web browser.
2. **Run Core Practical:**
3. Navigate to `PRACTICAL/index.html`.
4. Double click or right click and choose **Open with Live Server** (or open in any browser).
5. Click **Dark Theme**, refresh the browser tab, and observe that the dark mode remains active.
6. Click **Clear Preference** to reset.
7. **Run Case Study:**
8. Navigate to `CASE STUDY/index.html`.
9. Open the file in your browser.
10. Click on any seminar topic (e.g., *XPath*, *XSL Transformations*, or *Introduction to XML*).
11. Notice the alert box, the table cell highlight, and the updated "Storage Info" card.

12. Open Developer Tools (`F12` -> **Application** / **Storage** tab -> **Local Storage** & **Session Storage**) to inspect the stored key-value pairs live.

🎯 Learning Outcomes

- Successfully utilized HTML5 Web Storage API (`localStorage` and `sessionStorage`) for persistent and session-based state management.
- Understood object serialization and deserialization using `JSON.stringify()` and `JSON.parse()` .
- Implemented real-time DOM updates and event handling coupled with browser storage.
- Built accessible, responsive user interfaces with dark mode toggles and semantic HTML tables.

👤 Developer Information

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